

Lahore AQI Predictor

Open-Meteo + Hopsworks + Machine Learning

Repository: github.com/Shanzay2000/AQI-Predictor-Dashboard

<https://aqi-predictor-dashboard-shanzay.streamlit.app/>

1. Project overview

I built an AQI forecasting system for Lahore, Pakistan which uses only the latest six months of data for training and predicts AQI for the current day and the next 72 hours. The main purpose was not only to train a model, but to build the full pipeline: collect data, create features, store them in Hopsworks, train and compare models, register the best model, automate the process, and show the results in a Streamlit dashboard.

The data comes from Open-Meteo and the weather variables include temperature, humidity, precipitation, pressure and wind speed. Air-quality variables include PM2.5, PM10, CO, NO2, SO2 and ozone. The historical AQI field is used as the prediction target. The Future AQI from Open-Meteo is not used as a model input.

2. System design

The project is split into a few simple parts. Open-Meteo provides the raw data -> A feature pipeline prepares the data and writes it to Hopsworks -> The training script reads the latest six-month window, compares several regression models and saves the best one. The dashboard loads that model and produces the live forecast.

3. Data and feature engineering

The initial backfill downloads roughly six months (limited at 6 months to prevent hitting the Hopsworks limit) of hourly Lahore data. The six-month boundary is recalculated when the pipeline runs, so the training set does not keep growing indefinitely. For example, a run on 6 September uses data starting around 6 March.

- **AQI history:** current AQI, 1h/3h/6h/24h lags, AQI change, and 6h/24h rolling averages.
- **Weather:** temperature, humidity, precipitation, pressure and wind speed.
- **Pollutants:** PM2.5, PM10, CO, NO2, SO2 and ozone.
- **Time:** hour, day of week and month, including cyclic time features.

Hopsworks stores the raw and engineered data in separate Feature Groups. A Feature View is used to read the labelled training data, and the selected model is stored in the Hopsworks Model Registry.

The correlation matrix below (Figure 1) shows how the engineered features relate to each other and to AQI. PM2.5 and PM10 are the most strongly correlated with AQI (0.58 and 0.47 respectively), which matches expectations since particulate matter drives the AQI calculation directly. NO2 and ozone show a strong negative relationship (-0.73), consistent with the photochemistry that consumes NO2 to form ozone.

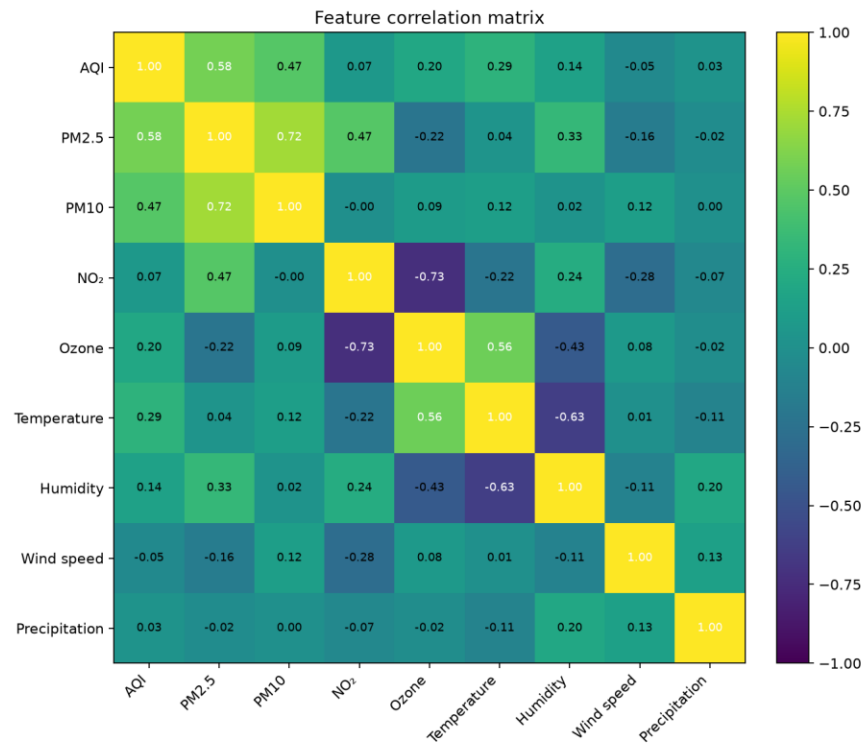


Figure 1. Feature correlation matrix.

4. Model training and results

The data is split chronologically rather than randomly: the oldest 70% is used for training, the next 15% for validation, and the newest 15% for testing. 70-15-15 split.

The pipeline compares Ridge Regression, Random Forest, Histogram Gradient Boosting, an MLP neural network, and XGBoost. The model with the lowest validation RMSE is selected and in the final run, XGBoost gave the lowest validation RMSE and was selected as the forecasting model.

Model	RMSE	MAE	R ²
XGBoost	3.9096	2.4290	0.9857
HistGradientBoosting	4.0736	2.4350	0.9845
Ridge	4.9175	2.9760	0.9774
Random Forest	5.6126	3.8578	0.9706
MLP	9.8428	7.4682	0.9094

XGBoost was the best model in the final comparison. It achieved a validation RMSE of 3.9096, MAE of 2.4290 and R² of 0.9857. Histogram Gradient Boosting was a close second with RMSE 4.0736. The strong results are partly because short-term AQI is closely related to recent AQI values and pollutant concentrations.

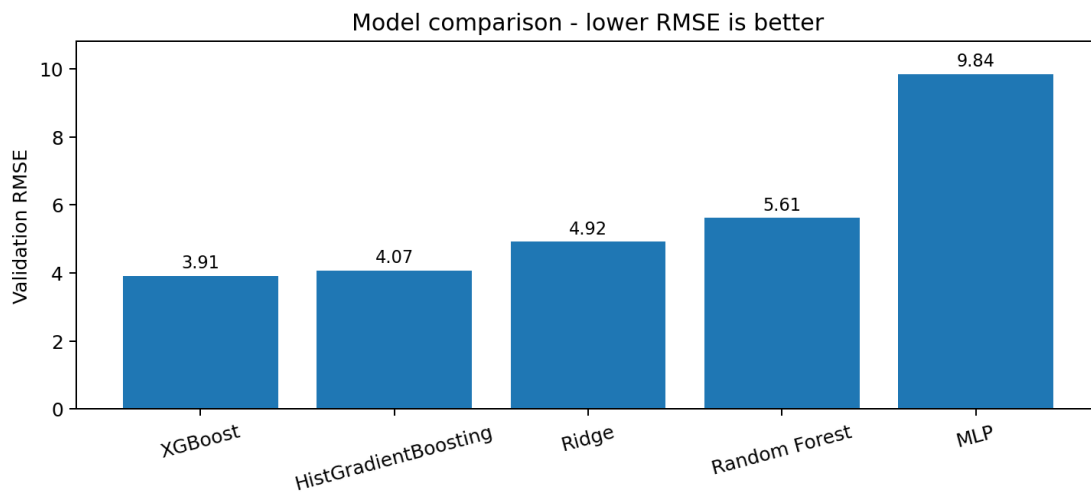


Figure 2. Validation RMSE from the final model comparison.

5. Forecasting

The retained forecasting approach is recursive. The best one-hour model predicts the next hour, that predicted AQI becomes part of the state for the following hour, and the process continues until 72 hourly values are produced. The Day 1, Day 2 and Day 3 values shown on the dashboard correspond to the +24h, +48h and +72h points on this same forecast path. A limitation of this approach is that errors can carry forward over longer predictions.

SHAP was added because explainability was an important part of the project brief. The dashboard has two types of explanation:

- Global SHAP shows which features generally matter most to the selected model.
- Local SHAP explains a particular forecast by showing which feature values pushed the prediction higher or lower.

The local explanation is useful for questions such as why the model predicted a high AQI on Day 2. It can show, for example, that PM2.5, recent AQI and low wind speed pushed the prediction upward. SHAP explains the model decision; it should not be interpreted as proof that a feature caused the pollution level.

6. Dashboard

The Streamlit dashboard shows the current Lahore AQI, the 72-hour hourly forecast, separate Day 1/Day 2/Day 3 values, six-month history, the selected model, evaluation results, an Actual-vs-Predicted plot, and SHAP explanations. AQI warning messages are also shown when the forecast enters unhealthy ranges.

Two GitHub Actions workflows were prepared: the feature pipeline runs hourly and the training pipeline runs daily. The Hopsworks API key and project name are stored as GitHub secrets instead of being committed to the repository. The public dashboard is intended to run on Streamlit Community Cloud.

7. Main issues faced

- GitHub actions does not run hourly.
- Project name had to be highly specific to work.

8. Limitations and conclusion

The main limitation is the recursive 72-hour forecast as the prediction error can accumulate as the predictions keep adding up hour by hour into the final prediction. I have also limited the data to past 6 months as to not run out of space on the free tier of Hopsworks but adding more data would improve the model. Environmental events such as crops/forest fires are difficult to predict.

To conclude, the project uses an external weather and pollution API, creates time-series features, stores them in Hopsworks, compares multiple models, registers the best model, supports hourly/daily automation and provides a Streamlit dashboard with AQI forecasts, evaluation plots and SHAP explanations. The strongest confirmed model result was XGBoost which achieved a validation RMSE of 3.9096, MAE of 2.4290 and R^2 of 0.9857.